- Rebuilt from my own research project, Play to Learn in VR (P01.01), continuing the student study from the Research Proposal, Definitions Practice, and Extended Codebook examples: one coder, 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit: the plan and the pilot as one document, the coded pilot data, and the code that computed the stability figure. The draw, the codes, and every figure are real: the tweets were drawn by the script shown, coded twice with the codebook, and the numbers are the scripts' own output. The minutes and the dates are a student's schedule.
- No tweet is quoted. Each case is paraphrased, and the data file carries tweet IDs and codes, not text, so a search cannot lead back to an account.
- The rubric map at the end shows which grading row each part earns and why.

The example

Sampling Plan and Pilot: Learning Talk, VR Tone, and the Balance of Benefits and Obstacles

Files: MC501-pilot-codes.csv (25 rows, one per tweet), 01-draw-sample.R, and 02-pilot-stability.R.

Part 1: Sampling plan

Population, frame, and the gap

The population my claims cover is English-language tweets, not retweets, posted from January 1 through December 31, 2021, that mention virtual reality (VR) together with the word play, in any sense, from any account.

The frame is the Play to Learn in VR analysis file, which holds the text of 6,108 tweets from a 2021 archive of 100,363 collected through the Twitter Academic Research API: all 3,054 archive tweets that contain one of 34 learning words, plus 3,054 drawn at random from the other 97,309.

The two differ in three places. First, coverage: a tweet the archive's search missed, or one deleted before collection, belongs to the population and can never be drawn, so my results describe tweets that survived into the archive. Second, composition: learning-word tweets are half the frame but 3 percent of the archive, so the frame is deliberately unrepresentative, and any whole-sample figure must be weighted back. Third, the text itself: the archive stripped digits and punctuation, which decision rule 3 handles but which can hide an emoji-free tone cue such as an exclamation mark.

Method

I use a stratified random sample with equal allocation: two strata, tweets with a learning word and tweets without one, and 75 drawn at random from each, seed 1106. Simple random sampling would yield about five learning-word tweets in 150, which cannot support H1's comparison or RQ1's count of what learning tweets name.

This design lets me estimate the share of each code within each stratum and compare the strata, and it lets me test H1 and RQ2 on the coded groups inside the sample. It does not let me estimate a single figure for all 2021 tweets, such as the share that are positive about VR, without weighting each stratum by its archive share, 3,054 against 97,309. It also does not fix the size of H1's groups: the strata are defined by a word, but H1 compares tweets I code as referring to learning, and the pilot shows those groups will be uneven.

Size: the justification

My justification is a resource limit, paired with a sensitivity analysis. The limit is coding time: as one coder I have about four hours, and the calibration below turns that into 150 tweets. A resource limit says nothing about what the study can detect, so I computed the smallest effect 150 tweets can find with 80 percent power at a .05 significance level. For H1, a two-by-three table (learning reference by tone), the answer is Cohen's w = .25; for RQ2, a two-by-four table (learning reference by the benefit-obstacle balance), w = .27. Both sit between Cohen's small (.10) and medium (.30) benchmarks.

That carries into how I will read the results. A significant H1 with w near .25 or larger is a finding of practical size. A null H1 does not show that learning talk and tone are unrelated; it shows only that any association is probably smaller than w = .25, and I will report it that way, with the confidence interval for Cramér's V.

Size: the time calibration

I timed the first 20 pilot tweets: 20 minutes, 60 seconds per tweet with four variables. Building the codebook on 30 immersion tweets and running this pilot took about 1.25 hours, leaving about 2.5 hours, or 9,000 seconds, for the full pass: 9,000 divided by 60 is 150 tweets. The nearest sizing-table row is short posts, captions, and headlines (20 to 60 words, 75 seconds, 150 items); my measured rate is faster, so 150 fits, and it is well above the floor of 50.

Date range and filters

Applied in this order in 01-draw-sample.R, so anyone running it gets the same tweets:

1. Posted from January 1 through December 31, 2021.
2. Language tag English.
3. Not flagged as possibly sensitive by Twitter.

4. Of tweets with identical text, only the lowest tweet ID is kept, since a bot reposting one promotion would otherwise count many times.
5. The 30 immersion tweets that built the codebook are removed.

Retweets were excluded at collection. Draws 76 to 90 in each stratum are a reserve list, used in order to replace any tweet marked out of frame.

Step	Tweets left	Removed
Frame: VR and play tweets with text	6,108	
Posted 01-01 through 12-31, 2021	6,108	0
Language tag English	6,105	3
Not flagged sensitive	5,833	272
First copy of repeated text only	5,236	597
Immersion tweets removed	5,206	30
Learning-word stratum / other stratum	2,359 / 2,847	
Drawn: sample / reserve, per stratum	75 / 15	

Part 2: Pilot

Items coded

I coded 25 tweets, the first 13 draws from the learning-word stratum and the first 12 from the other, with the codebook frozen on 11-02. MC501-pilot-codes.csv has one row per tweet: ID, stratum, draw order, date, my four codes from each pass, and the like count, looked up only after coding.

The stability figure

This is intracoder reliability, or stability: both passes are mine, so the figure measures how consistently I apply the codebook over time, not agreement between coders. I coded all 25 tweets on 11-03 and recoded them on 11-06, 72 hours later, from a fresh sheet of IDs and text in a shuffled order with no code columns, so my first codes stayed hidden. 02-pilot-stability.R joins the passes and computes Krippendorff's alpha for each variable with v2v::reliability(), and the ordinal alpha for tone with irr::kripp.alpha().

Variable	Passes agree	Krippendorff's alpha
Learning reference	25 of 25 (100%)	1.00
Tone toward VR	24 of 25 (96%)	.93 nominal, .94 ordinal
Affordance named	24 of 25 (96%)	.94
Constraint named	24 of 25 (96%)	.89

Every variable clears the .80 threshold in my protocol. The alpha for learning reference is a ceiling on 25 tweets, not proof the variable is easy, and it will be retested on the fresh

reliability sample after the full pass. Each of the three disagreements was a tweet no rule covered; each produced a revision.

Edge cases

Case (paraphrased)	Decision	Rule
Four tweets use play in another sense: technology plays a role in training, in retail, in students' experience, or in isolating people.	In frame, coded as usual.	Decision rule 1 (revision A).
A sword game sends a player to a real sword art; a teacher plans a body-awareness lesson without VR.	Learning reference 0.	New learning reference rule 7 (revision B).
A player says they learned not to play VR right after waking up.	Learning reference 0 (an idiom); constraint 0 (a problem hinted at, not named).	Learning reference rule 1 and new constraint rule 5 (revisions C and F).
A writer blames VR for cutting people off from the world. Passes split, constraint 0 then 4.	Constraint 4.	New constraint rule 6 (revision F).
A queue-simulator promotion calls its new levels bug-free. Passes split, tone 3 then 2.	Tone 2: a claim about the product, not a judgment of the experience.	Tone rule 2 (revision D).
VR is said to give students with disabilities a safe, immersive experience. Passes split, affordance 5 then 1.	Affordance 5: the benefit is who gets to take part.	Affordance category 5 and new rule 5 (revision E).
A player keen on VR has no headset yet and expects VR to spread as it gets cheaper.	Constraint 2, since price is stated.	Existing constraint rule 4.
A podcast invites listeners to learn how VR is changing work.	Learning reference 0.	Existing learning reference rule 2.
Game names and a German contest word, tagged English.	Coded as a hashtag string.	Existing decision rule 4.
A cryptocurrency metaverse promotion offers education, earnings, games, and social networks.	Affordance 5, earnings, the longest.	Existing tie rule in the affordance definition.

Revisions the pilot forced

Revision	Before	After
A. Decision rule 1	Before coding, check that the tweet is about virtual reality. If VR appears only inside another word, a username, or an unrelated abbreviation, and the tweet is about something else, mark it out of frame, code nothing, and replace it with the next tweet on the reserve list.	The same rule, plus: The word play may carry any sense. A tweet saying VR will play a role in training is in frame and is coded like any other.
B. Learning reference	(No rule 7.)	7. Learning that VR prompts but that happens outside VR (a sword game that sends a player to a real sword class, a lesson planned without VR) is coded 0.
C. Learning reference rule 1	... part of an idiom (hype train, learned the hard way, learned my lesson) is coded 0 unless another clause describes learning.	... part of an idiom (hype train, learned the hard way, learned my lesson, today I learned) is coded 0 unless another clause describes learning.
D. Tone rule 2	An advertisement that lists features or a release date with no evaluative word is coded 2.	An advertisement that lists features, a release date, or a quality claim about the product itself (bug-free, new levels, sharper graphics) is coded 2. Only a word that judges the experience (amazing, fun, the best) makes it 3.
E. Affordance named	5 = other benefit. (No rule 5.)	5 = other benefit (including access for people who could not otherwise take part). Rule 5: When VR is said to open an experience to a named group who would otherwise miss it, code 5, even if the experience is also called immersive.
F. Constraint named	(No rules 5 or 6.)	5. A problem the tweet hints at without naming is coded 0. 6. A harm the writer says VR does to people (cutting them off from others, blurring fantasy and reality) is coded 4, because it gives a reason to use VR less.

What the pilot says about the full run

At 60 seconds a tweet, the 125 new tweets plus a recode of the 25 pilot tweets take 2.5 hours, which I will split into four sittings of under 40 minutes so tone does not drift late in a session, then 15 minutes to recode the fresh reliability sample after 48 hours. Two things change. I freeze the six revisions before coding anything new. And the pilot coded only 7 of 13 learning-word tweets, and none of the others, as referring to learning, so H1's groups will be near 40 and 110, not 75 and 75; the RQ2 balance had only 2 of 25 tweets in its both-benefit-and-obstacle group, so if any expected cell in that table falls below 5, I will report Fisher's exact test beside the chi-square.

The stability code and its output

02-pilot-stability.R, as run, with its output:

```
options(scipen = 999)
library(tidyverse)
library(v2v)

pilot <- read_csv(
  "MC501-pilot-codes.csv",
  col_types = cols(tweet_id = col_character())
)

pilot %>%
  summarize(
    items = n(),
    learning = mean(learning_p1 == learning_p2),
    tone = mean(tone_p1 == tone_p2),
    affordance = mean(affordance_p1 == affordance_p2),
    constraint = mean(constraint_p1 == constraint_p2)
  ) %>%
  print()
# A tibble: 1 x 5
#   items learning  tone affordance constraint
#   <int>    <dbl> <dbl>      <dbl>      <dbl>
# 1    25        1  0.96       0.96       0.96

tibble(
  variable = c("learning", "tone", "affordance", "constraint"),
  alpha = c(
    reliability(pilot$learning_p1, pilot$learning_p2, method =
"alpha")$statistic,
    reliability(pilot$tone_p1, pilot$tone_p2, method = "alpha")$statistic,
    reliability(pilot$affordance_p1, pilot$affordance_p2, method =
```

```
"alpha")$statistic,
    reliability(pilot$constraint_p1, pilot$constraint_p2, method =
"alpha")$statistic
  )
) %>%
  mutate(alpha = round(alpha, 3)) %>%
  print()
# A tibble: 4 x 2
#   variable   alpha
#   <chr>      <dbl>
# 1 learning   1
# 2 tone       0.928
# 3 affordance 0.935
# 4 constraint 0.89

irr::kripp.alpha(rbind(pilot$tone_p1, pilot$tone_p2), method =
"ordinal")$value %>%
  round(3)
# [1] 0.935

pwr::pwr.chisq.test(N = 150, df = 2, sig.level = 0.05, power = 0.80)
#               w = 0.2534384
#               N = 150
#              df = 2
#       sig.level = 0.05
#           power = 0.8
pwr::pwr.chisq.test(N = 150, df = 3, sig.level = 0.05, power = 0.80)
#               w = 0.2695969
#               N = 150
#              df = 3
#       sig.level = 0.05
#           power = 0.8
```

Rubric map

The plan rows total 25, size justification 15, pilot data and the reliability computation 20, and edge cases, revisions, and the full run 15.

Rubric row	Points	Where the example earns it	Why it earns the full row

Rubric row	Points	Where the example earns it	Why it earns the full row
Plan: population, frame, and the gap	12	Population, frame, and the gap	It defines the population, names the actual file drawn from, and locates three specific differences (coverage, composition, the stripped text), each with its consequence, instead of assuming the two match.
Plan: method	13	Method	It names and justifies stratified random sampling with equal allocation, then says what the design lets the student estimate (within-stratum shares, comparisons, the tests) and what it does not (a weighted whole-population figure, fixed group sizes for H1).
Size: the justification	10	Size: the justification	It names one of the three justifications (a resource limit paired with a sensitivity analysis), gives the detectable effect for each planned test, and carries it into how a null result will be reported.
Size: the time calibration	5	Size: the time calibration	All three parts: a rate measured on a timed twenty-item trial, the corpus size that rate produces, and the nearest sizing-table row.
Pilot: items coded	8	Items coded, and MC501-pilot-codes.csv	Twenty-five items, inside twenty to thirty, coded and submitted as data with one row per item.
Pilot: the stability figure	12	The stability figure, its table, and the stability code	Alpha from recoding the student's own pilot items 72 hours later with the first codes hidden, computed by code that is submitted, and named as stability, not agreement between coders.
Edge cases	6	The edge-case table	Every case met in the pilot, each with the decision and the rule added, including the four that an existing rule settled.
Revisions the pilot forced	6	The revisions table	Before and after wording for each of the six changes.
What the pilot says about the full run	3	What the pilot says about the full run	One paragraph with both parts: how long the full run will take, and what will change because of the pilot.

Length: about 1,010 words of the student's own writing across the plan and the pilot, not counting headings, tables, or code, inside the assignment's range, plus 25 coded items in the CSV.